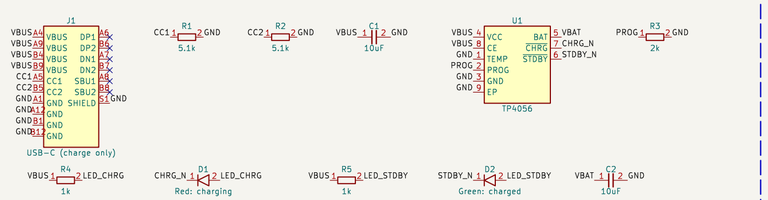
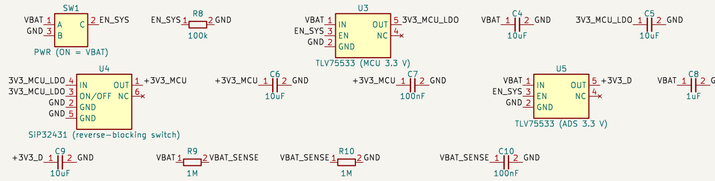
1 USB-C charging (TP4056, 580 mA)



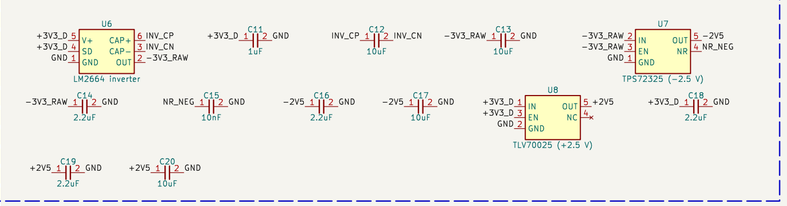
2 LiPo cell and protection (DW01A + FS8205A)



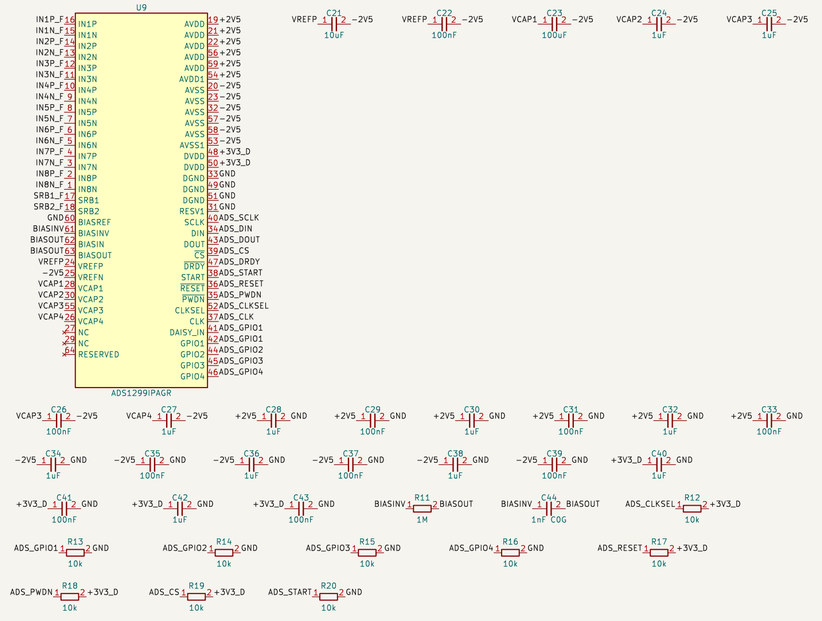
3 Power switch and 3.3 V rails (MCU rail is reverse-blocked)



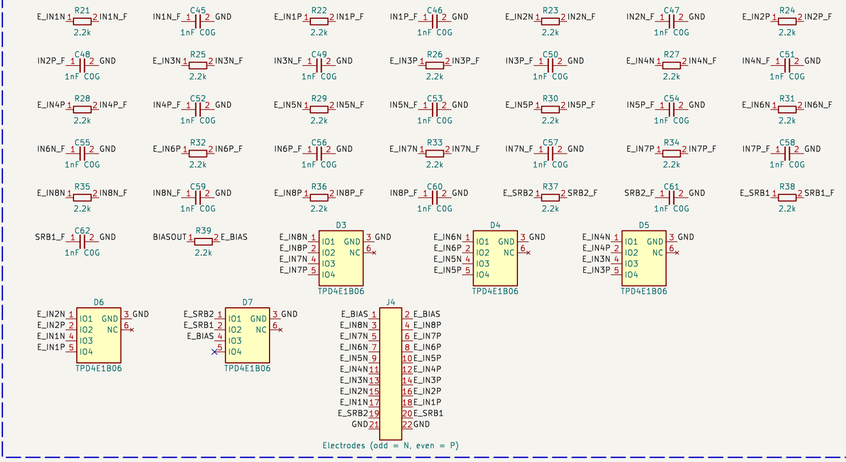
4 +2.5 V / −2.5 V analog supplies (OpenBCI Cyton circuit)



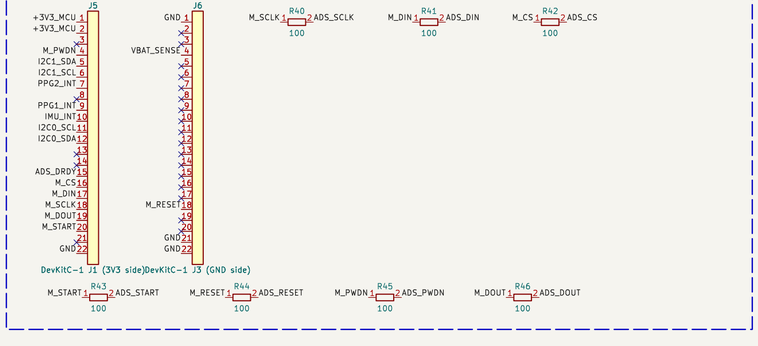
5 ADS1299 analog front end



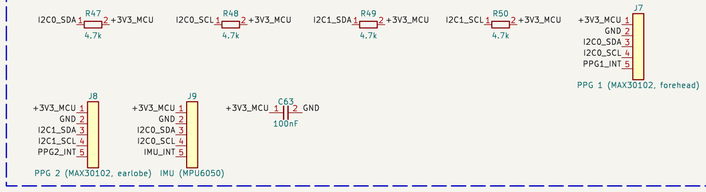
6 Electrode inputs: ESD clamp, 2.2k series, 1 nF C0G to GND



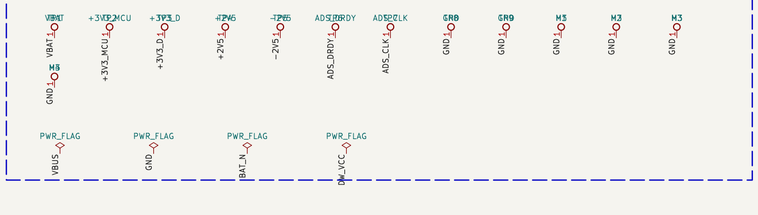
7 ESP32-S3-DevKitC-1 sockets and SPI series resistors



8 Sensor ports (I2C)



9 Test points and mounting holes



Safety: battery only while electrodes are on a person. USB-C is for charging with nobody connected.
To flash firmware: PWR switch ON, nobody connected to the electrodes.
Electrode convention (Cyton): electrodes on N inputs, reference on SRB2, BIAS to the bias electrode.

Generated from gen/design.py: edit the design file, then regenerate
Research prototype, not a medical device. Battery only while electrodes are on a person.
8-ch ADS1299 EEG + ESP32-S3 devkit. Derived from OpenBCI Cyton (CC BY-SA 4.0)

Neuryo

Sheet: /
File: neuryo.kicad_sch

Title: Neuryo EEG board

Size: A2	Date: 2026-10-30	**Rev: A**
KiCad E.D.A. 8.0.9		Id: 1/1